MATRIX ADDITION AND MULTIPLICATION

LINEAR ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Matrix

Adding Matrices

Multiplying Matrices

- Scalar and Vector

MATRIX

A **rectangular array** of numbers, symbols, or expressions, arranged in **rows** and **columns**

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ . & . & \dots & . \\ . & . & \dots & . \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

DIMENSION

The dimension of a matrix is the number of rows and columns

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ . & . & \cdots & . \\ . & . & \cdots & . \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$m \times n$

DIMENSION

The dimension of a matrix is the number of rows and columns

$$\boldsymbol{m \times n}$$

$$\boldsymbol{A = \begin{bmatrix} 5 & -3 & 0 \\ 1 & 9 & -4 \end{bmatrix}}$$

2×3 Matrix

$$\boldsymbol{B = \begin{bmatrix} 5 & 2 \\ 1 & 0 \\ -4 & 8 \end{bmatrix}}$$

3×2 Matrix

ELEMENTS

$$A = \begin{bmatrix} 5 & -3 & 0 \\ 1 & 9 & -4 \end{bmatrix}$$

The elements of a matrix are the **entries**

a_{ij}

$$B = \begin{bmatrix} 5 & 2 \\ 1 & 0 \\ -4 & 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 5 & -3 & 0 \\ 1 & 9 & -4 \end{bmatrix}$$

The elements of a matrix are the **entries**

a_{ij}

$a_{11} = 5$ $\quad a_{12} = -3$ $\quad a_{13} = 0$

$a_{21} = 1$ $\quad a_{22} = 9$ $\quad a_{23} = -4$

$$B = \begin{bmatrix} 5 & 2 \\ 1 & 0 \\ -4 & 8 \end{bmatrix}$$

The elements of a matrix are the **entries**

a_{ij}

$b_{11} = 5$ $\quad$ $b_{12} = 2$

$b_{21} = 1$ $\quad$ $b_{22} = 0$

$b_{31} = -4$ $\quad$ $b_{32} = 8$

ADDING MATRICES

Same Dimensions

Matrices must have the same number of rows and columns

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

Element-wise Addition

Adding elements in the **same position** from both matrices

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

ADDING MATRICES

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$A + B = \begin{bmatrix} \quad & \quad & \quad \\ \quad & \quad & \quad \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

SUBTRACTING MATRICES

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$A - B = \begin{bmatrix} \quad & \quad & \quad \\ \quad & \quad & \quad \end{bmatrix}$$

Matrices with different dimensions cannot be added or subtracted

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

PROPERTIES OF MATRIX ADDITION

Addition is associative

$$(\boldsymbol{A}+\boldsymbol{B})+\boldsymbol{C}=\boldsymbol{A}+(\boldsymbol{B}+\boldsymbol{C})$$

Additive identity (the zero matrix)

$$\boldsymbol{A}+[\mathbf{0}]=\boldsymbol{A}$$

Addition is commutative

$$\boldsymbol{A}+\boldsymbol{B}=\boldsymbol{B}+\boldsymbol{A}$$

MULTIPLYING MATRICES

SCALAR MULTIPLICATION

$$3 \cdot \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

Scalar is a **single** numerical value

$$\begin{bmatrix} & & \\ & & \end{bmatrix}$$

PROPERTIES OF SCALAR MULTIPLICATION

Scalar multiplication is distributive

$$r(A + B) = rA + rB$$

Scalar multiplication is associative

$$(r \cdot l)A = r(l \cdot A)$$

Multiplication by 0

$$\mathbf{0} \cdot A = [\mathbf{0}]$$

Scalar multiplication is commutative

$$r \cdot A = A \cdot r$$

ROW VECTOR

A matrix that has only **one row**

$$\vec{w} = [w_1 \quad w_2 \quad w_3 \quad \dots \quad w_n]$$

$\mathbf{1 \times n}$ **matrix**

ROW VECTOR

A matrix that has only one row

$$\vec{w} = \begin{bmatrix} 3 & 6 & 1 & -1 \end{bmatrix}$$

1×4 matrix

COLUMN VECTOR

A matrix that has only **one column**

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ . \\ . \\ v_n \end{bmatrix}$$

$\boldsymbol{n \times 1}$ **matrix**

COLUMN VECTOR

A matrix that has only one column

$$\vec{v} = \begin{bmatrix} 1 \\ -2 \\ 3 \\ 5 \end{bmatrix}$$

4×1 matrix

VECTOR MULTIPLICATION

Multiplying a $1 \times n$ **row vector** with an $n \times 1$ **column vector**

$$\vec{w} = \begin{vmatrix} 3 & 6 & 1 & -1 \end{vmatrix} \qquad \vec{v} = \begin{vmatrix} 1 \\ -2 \\ 3 \\ 5 \end{vmatrix}$$

$$\sum_{i=1}^{n} \vec{w}_i \vec{v}_i$$

$$\vec{w} \cdot \vec{v} =$$

$$\underset{m \times r}{\overset{A}{\begin{bmatrix} \\ i \\ \\ \end{bmatrix}}} \cdot \underset{r \times n}{\overset{B}{\begin{bmatrix} & j & \\ & & \\ & & \end{bmatrix}}} = \underset{m \times n}{\overset{C}{\begin{bmatrix} & & \\ & c_{ij} & \\ & & \end{bmatrix}}}$$

columns of A = **rows** of B

dot product of *i-th* row of A and *j-th* column of B

$$\underset{4\times 2}{\overset{A}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \underset{2\times 3}{\overset{B}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \underset{4\times 3}{\overset{C}{\begin{bmatrix} & & \\ & & \\ & & \\ & & \end{bmatrix}}}$$

$$\overset{B}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}} \cdot \overset{A}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}} = \overset{C}{\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}}$$

2×3 $\qquad$ 4×2 $\qquad$ **invalid**

columns of A = **rows** of B

EXERCISE

For $\boldsymbol{Y} = \begin{bmatrix} \mathbf{1} & \mathbf{5} \\ \mathbf{7} & \mathbf{2} \end{bmatrix}$ and $\boldsymbol{Z} = \begin{bmatrix} \mathbf{-1} & \mathbf{3} \\ \mathbf{9} & \mathbf{5} \end{bmatrix}$

Find (a) $\boldsymbol{YZ}$ and $(b)\boldsymbol{ZY}$

IDENTITY MATRIX

A **square matrix** in which all the elements of the principal **diagonal** are **ones**, and all other elements are zeros

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

For $\boldsymbol{I} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$ and $\boldsymbol{A} = \begin{bmatrix} \mathbf{3} & \mathbf{-4} & \mathbf{7} \\ \mathbf{8} & \mathbf{9} & \mathbf{1} \\ \mathbf{-2} & \mathbf{5} & \mathbf{6} \end{bmatrix}$

Find (a) $\boldsymbol{IA}$ and $(b)\boldsymbol{AI}$

PROPERTIES OF MATRIX MULTIPLICATION

	Numbers	Matrices
Associative Property	$(ab)c = a(bc)$	$(AB)C = A(BC)$
Commutative Property	$ab = bc$	$AB \neq BC$
Distributive Property	$a(b + c) = ab + ac$	$A(B + C) = AB + AC$
Multiplicative Identity	$1 \cdot a = a$	$I \cdot A = A$

SEATWORK